Arpita Balaso Karande

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Education

B.Tech. in Information Technology [Third Year] — Shivaji University Aug. 2023 – May 2027
Government College of Engineering, Karad

Projects

1. **Hand Gesture Recognition System for Communication**
   - Engineered a real-time system using **Python, OpenCV, and MediaPipe** to translate hand gestures into spoken commands.
   - Implemented custom classification logic to detect four distinct gestures and convert them to speech using **pyttsx3 (Text-to-Speech).**
   - Integrated the **Twilio API** for remote communication, enabling the system to automatically send critical SMS alerts based on recognized gestures.
2. **Intelligent Traffic Signal Automation**
   - Developed an autonomous traffic management solution using **sensors** and **microcontrollers** to dynamically optimize traffic flow.
   - Implemented a mechanism for **real-time detection** of vehicle density to intelligently adjust signal timings.
   - The primary goal was to **mitigate congestion, lower fuel consumption, and reduce waiting periods** at busy intersections.
   - Gained essential practical experience in core concepts of **IoT**, **embedded systems**, and **automation**.
3. **Smart Crop Advisory System (AI-Driven)**
   - Developed a profit-aware **Crop Advisor** using **Random Forest** and **Classical Machine Learning model** models for yield and price prediction.
   - Engineered high-impact features from two large datasets to score crops based on **Profit** and **Sustainability**.
   - Designed a deployable architecture using **FastAPI** (API Gateway) and few other **Model Services** to deliver multilingual, risk-based advisories.

Technical Skills

- **Programming:** C, C++, Python, SQL
- **Web/Frameworks:** React Native, Flask, Node.js, FastAPI
- **ML/Data:** OpenCV, MediaPipe, Random Forest, Pandas, NumPy, AI/ML Basics
- **Tools/DevOps:** Git/GitHub, Twilio API, Microcontrollers, Sensors

Internships

Python Programming Intern | IFS India Mercantile Private Limited | Jun 2024 – Jul 2024 *Pune, Maharashtra, India*

- Successfully completed various project modules **independently**, developing foundational skills in **Python programming** for professional projects.
- Demonstrated a **strong independent work ethic** and professional project delivery by completing all assignments within corporate timelines.
- Gained valuable experience working effectively in a **remote setting** while adhering to corporate standards and objectives.

Project Intern | Deltiin India Tech Private Limited | Jun 2024 – Aug 2024 *Karad, Maharashtra, India*

- Mastered **React.js** for front-end development and adhered to modern development standards.
- Gained hands-on experience with industry-standard tools, including **GitLab** for version control and **ASANA** for project management.
- Demonstrated ability to work in a **structured**, **professional setting** and adhere to project timelines over a two-month period.

Certifications

- Student Innovation Ambassador (IA) Training 'Foundation Level' (30 contact hours) | MoE's Innovation Cell & AICTE | Sept 2024
- C Programming Certification (Score: 90.00%) | Spoken Tutorial Project, IIT Bombay | Nov 2024
- C++ Programming Certification (Score: 80.00%) | Spoken Tutorial Project, IIT Bombay | Nov 2024

Organizational & Project Leadership

- SPOC Committee Member, GCEK JobFair'25 (Jan, 2025)
    - Played a core role in coordinating the college's Job Fair, serving as a **Single Point of Contact (SPOC)** for organizing teams & companies, and managing key logistical tasks.
- Core Team Member, New Building Inauguration (Oct, 2024)
    - **Entrusted with core responsibilities** for the inauguration of new college facilities, showcasing administrative reliability and high-level event coordination skills.
- Students Ambassador (MoE's) (Sep, 2024)
    - Championed **innovation and technology-driven initiatives** at institutional and national levels.
    - **Collaborated with peers and faculty** to design and implement programs fostering skill development and innovation.

Achievements

- Technical & Innovation Distinctions
    - Secured 2nd Place in the Science Day Exhibition Competition.
    - Successfully participated in 3 SIH (Smart India Hackathon) Internal Hackathons (from 2023 to 2025).
- Sports & Athletic Achievements
    - Achieved 1st Place in the Zonals Women Table Tennis Tournament for two consecutive years.
    - Secured 4th Place in the Inter-zonals Women Table Tennis Tournament for two consecutive years.

Technical Strengths

Analytical Thinking | Innovation | Problem Solving | Data Modeling | Real-Time Processing | ML Systems Design | Leadership | Organizational Management | Collaboration | Communication | Adaptability | Event Coordination

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